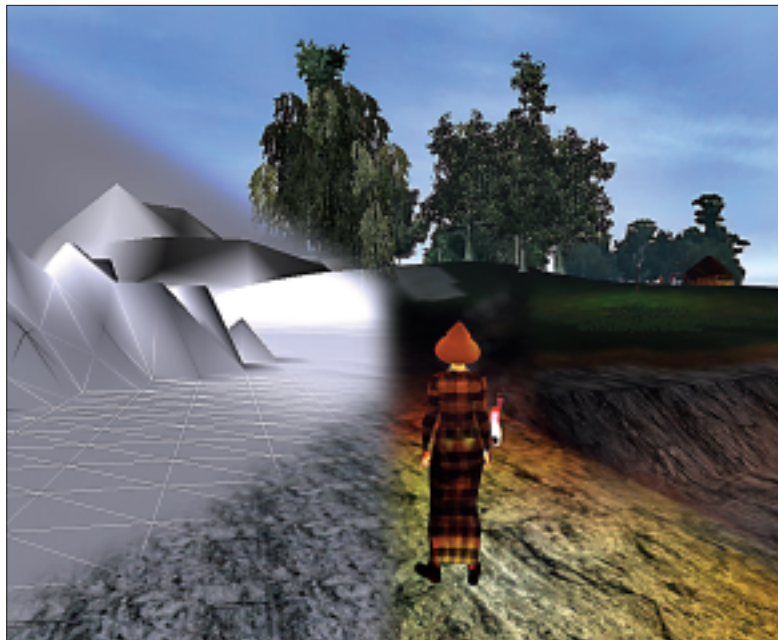




This partially rendered image shows how simple geometry can be enhanced by the use of texture and bump maps

spheres, cubes, cylinders, pyramids, etc. These can then be combined together using Boolean operations, either additively or subtractively. An example: the gear-wheel will need a hole drilled in the centre for it to fit on a shaft. After the gear-wheel has been created, you then create an appropriately sized cylinder, position it in the centre of the gear-wheel, and subtract it from the gear with a Boolean operation. Again, this is done automatically, many new triangles are created and the gear wheel now has the appropriate hole – the two objects have been replaced by just one. Building up complex structures in this way tends rapidly to create large numbers of triangles, many of which are unnecessary. It is often useful to go in and do some low-level editing, and many applications provide some type of simplification or tidying-up procedure. The gear-wheel now needs some colour.

There are usually many factors associated with a surface – even with individual triangles – that give the final look when it is rendered. The most obvious is colour, but this is not as simple as one might at first think. The colour of an object when viewed depends on the light that is reflected from its surface. The process of colouring an object during rendering is called shading. The reason for this term is simple: consider a ball illuminated by a light on only one side. The ball will not have uniform colour all around, but the side facing the light will be bright, and the side away from the light will be dark, be in shadow. To render this properly, the colour needs to be shaded around the ball, from bright on one side to dark on the other. There

are many different software techniques for shading, which we do not need to go into here.

Also, the colour of the light that hits the object in the first place may itself not be pure white, and the two main ways in which objects reflect that light are referred to as diffuse and specular. Think of a piece of brass for our gear-wheel or other shiny metal. The general colour of the brass is due to diffuse reflection, but the polished metal also reflects similarly to a mirror – this will give rise, when lights and bright objects are present, to specular highlights visible on the brass surface. There is more to colour than this, but those are the basic principles.

Next, consider a polished piece of wood. This will have a general diffuse colour, possible specular highlights if it is well polished, and also a pattern due to the grain of the wood. The pattern is often produced in modelling by using a raster image – in this case, perhaps even a digital photograph of real wood. Such an image is called a texture map, and within the software it needs to be specified just how the image is applied to the object surface. For example, with a building you might well have a texture map of a few bricks, perhaps a square section, and this will need to be repeated – tiled – over the surface many times. In another example, the airline logo on the side of an aeroplane needs to be used just once, with a particular relative size and exact position.

This process of applying the image to the object – and there are many variations possible – is called mapping, and it is worth spending time to study all the methods and try each of them out when learning modelling. This process of applying textures and other maps is one of the key techniques in creating realistic models. Patience and attention to detail are vital, and will be well rewarded.

The grain in a piece of wood as just described is simply a texture map, but there are many other types of map that can be used. For example, the surface of a piece of gold will reflect objects behind the camera (the viewpoint from which the image is rendered). These reflections can be created by techniques such as ray-tracing – described later – or by the use of a reflection map. The software will add the reflected image to the main texture of the material from the reflection map.

There are many other types of map used, and the latest version of MAX provides for a total 19. But perhaps the next most important is the bump map. Here, the colours in the map are interpreted by the software as either bumps or indentations in the surface. The coordinates of the vertices are not changed by the map – there are of course other tech-